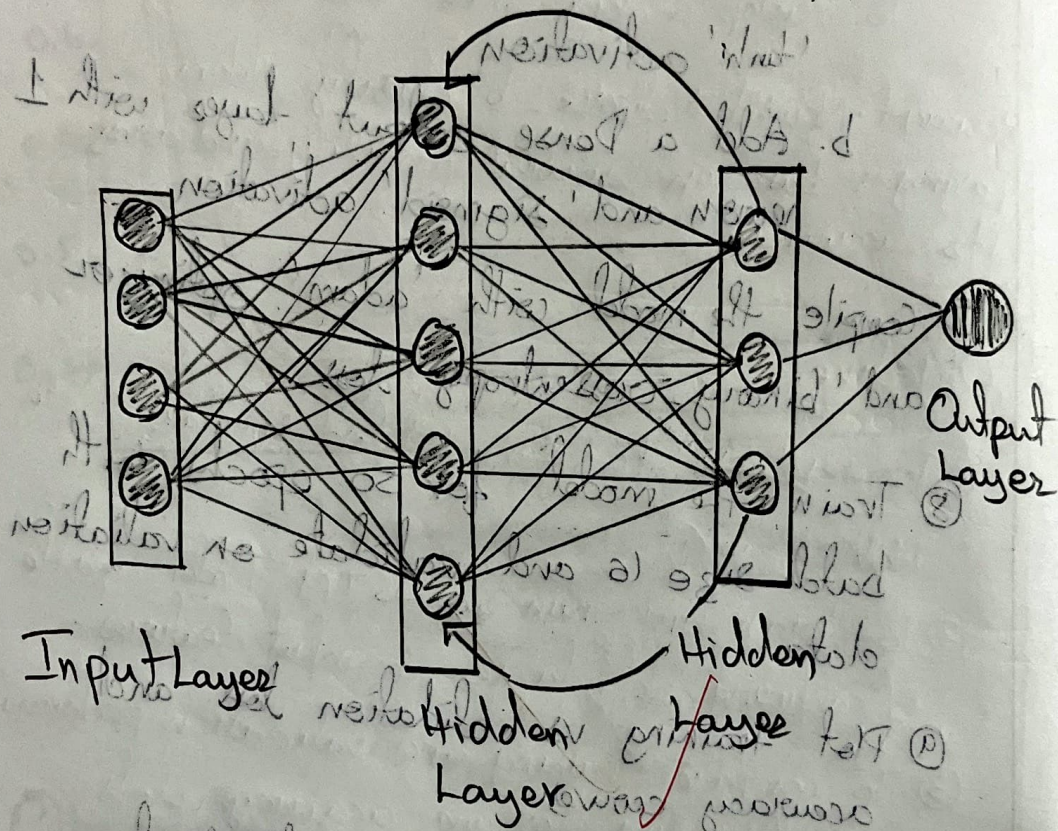


the variables has primary objective type
 before learning a model
 have this as input and output



Recurrent Neural Network

Result :-
 The code executed successfully
 with an accuracy of 80.42% and
 the loss vs Epoch curve was observed.

2-10-25
Exp. 9

Recurrent Neural Network

Aim:-

To build and train a Recurrent neural Network to predict passengers survival on the Titanic dataset

Objective:-

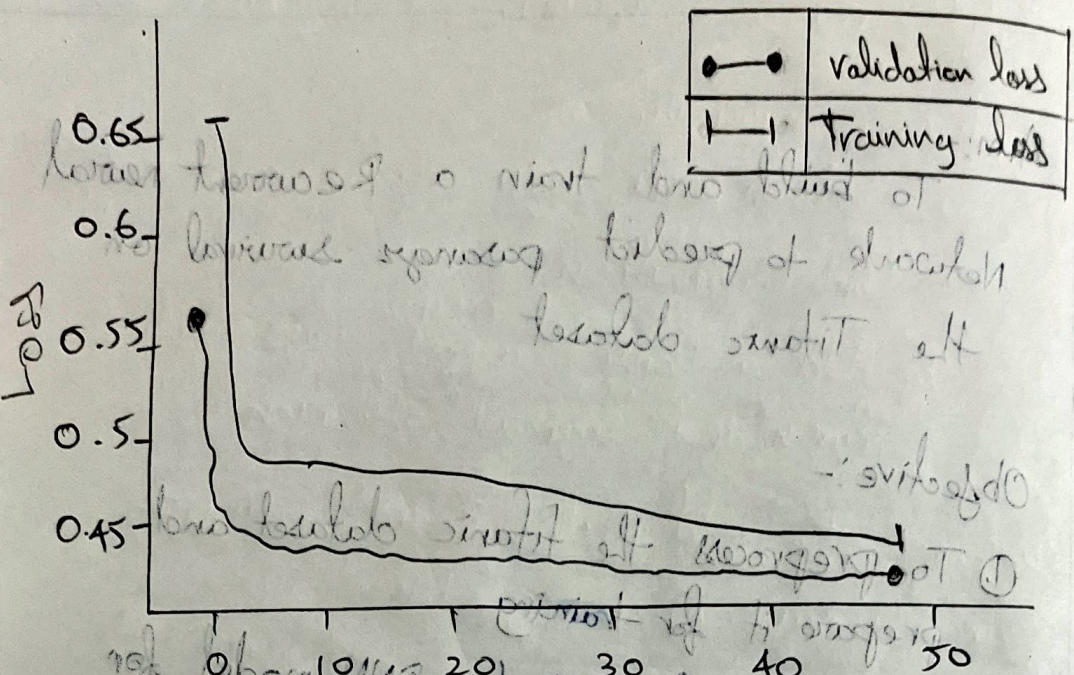
- ① To preprocess the titanic dataset and prepare it for training
- ② To implement a simple RNN model for binary classification
- ③ To visualize training and validation loss and accuracy curves
- ④ To evaluate the model performance using validation accuracy.

PseudoCode :

- ① Load the titanic dataset
- ② Select relevant features : Pclass, Sex, age and fare
- ③ Handle missing values and encode categorical features
- ④ Normalize the features using StandardScaler.
- ⑤ Reshape the input to 3D format suitable for RNN: (Samples, timestep, features)

Loss Function

25/07/2020



Epochs

Observation:-

Epoch 10, Loss : 0.4602

Epoch 20, Loss : 0.4553

Epoch 30, Loss : 0.4514

Epoch 40, Loss : 0.4468

Epoch 50, Loss : 0.4411

RNN Accuracy:-

79.89 %

for RNN (LSTM, GRU, etc.)

⑥ Split the dataset into training and validation sets

⑦ Define a sequential model:

a. Add a Simple RNN layer with 32 units and 'tanh' activation.

b. Add a Dense output layer with 1 neuron and 'sigmoid' activation

⑧ Compile the model using 'adam' optimizer and 'binary_crossentropy' loss.

⑨ Train the model for 50 epochs with batch size 16.

⑩ plot training and validation loss and accuracy curves

⑪ Evaluate the model on validation data and print accuracy

Result:

The code executed

successfully. and Loss vs Epoch graph was obtained

~~2/11/23~~
2/11/23